

as its first file. To find the second file for the new cabinet, once again take the first file in the left and right cabinets, compare them, and insert whichever is lower into the last position in the new cabinet. When either the left cabinet or the right cabinet is empty, take the remaining files in the non-empty cabinet and place them all together at the end of the new cabinet. After this, your new cabinet will contain all the files from the left and right cabinets, sorted in order. We have successfully merged our original two cabinets.

In Python, we'll use the variables `left` and `right` to refer to our original sorted cabinets, and we'll define a `newcabinet` list, which will start empty and eventually contain all elements of both `left` and `right`, in order.

```
newcabinet = []
```

We'll define example cabinets that we'll call `left` and `right`:

```
left = [1,3,4,4,5,7,8,9]
right = [2,4,6,7,8,8,10,12,13,14]
```

To compare the respective first elements of our left and right cabinets, we'll use the following `if` statements (which won't be ready to run until we fill in the `--snip--` sections):

```
if left[0] > right[0]:
    --snip--
elif left[0] <= right[0]:
    --snip--
```

Remember that if the first element of the left cabinet is lower than the first element of the right cabinet, we want to pop that element out of the left cabinet and insert it into the `newcabinet`, and vice versa. We can accomplish that by using Python's built-in `pop()` function, inserting it into our `if` statements as follows:

```
if left[0] > right[0]:
    to_insert = right.pop(0)
    newcabinet.append(to_insert)
elif left[0] <= right[0]:
    to_insert = left.pop(0)
    newcabinet.append(to_insert)
```
